

# **MZ UCA Cloud Academy**

## **AWS Solutions Architect Associate (SAA)**

### **Architecture Defense**

#### **AWS Organizations, Service Control Policies (SCPs), Landing Zones & Multi-Account Strategy**

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**Cloud Governance Architect**

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### **PURPOSE OF THE ARCHITECTURE DEFENSE**

In the real world, architects do not simply build solutions.

They defend their decisions.

Senior architects are expected to justify:

- ✓ Security decisions
  - ✓ Governance controls
  - ✓ Cost management approaches
  - ✓ Compliance strategies
  - ✓ Risk mitigation measures
  - ✓ Scalability plans
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## Your Mission

You are the Lead Cloud Governance Architect for a rapidly growing enterprise.

You must evaluate the current environment and defend your proposed architecture before an Executive Review Board.

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## BUSINESS CASE

### Ubuntu Retail Group (URG)

URG is expanding across Southern Africa and migrating critical business systems to AWS.

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#### Business Requirements

- ✓ Secure cloud environment
  - ✓ Regulatory compliance
  - ✓ Cost transparency
  - ✓ High availability
  - ✓ Centralized governance
  - ✓ Scalable cloud operating model
  - ✓ Enterprise-grade security
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#### Current Environment

Production

Development

Testing

Security

Finance

All operate within a single AWS account.

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### **Executive Concern**

"Can this architecture support long-term growth safely?"

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## **DEFENSE ROUND 1**

# **SHOULD WE USE AWS ORGANIZATIONS?**

### **Executive Question**

"Our engineers already have an AWS account.

Why do we need AWS Organizations?"

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### **Your Defense**

Explain:

1. What AWS Organizations provides.

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1. What governance problems it solves.

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1. What risks exist if we remain on a single account.

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### **Architect Recommendation**

☐ Implement AWS Organizations

☐ Remain Single Account

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**Defend Your Choice**

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**◆ DEFENSE ROUND 2**

**DESIGNING ORGANIZATIONAL UNITS**

**Executive Question**

"Why can't we simply place all accounts together?"

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**Task**

Design your Organizational Unit structure.

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AWS Organization



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### **Defense Questions**

Why was each OU created?

How does it improve governance?

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### **Architecture Principle**

Structure should reflect governance requirements.

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## **DEFENSE ROUND 3**

# **SCP GOVERNANCE STRATEGY**

### **Executive Question**

"Why do we need Service Control Policies if IAM already exists?"

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### **Explain**

Difference between:

IAM Policies

and

SCPs

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IAM Policies:

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SCPs:

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### **Governance Defense**

Provide three examples where SCPs protect the organization.

1.

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2.

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3.

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### **Critical Principle**

SCPs establish organizational guardrails.

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## **DEFENSE ROUND 4**

# **LANDING ZONE INVESTMENT**

### **Executive Question**

"Why should we spend time building a Landing Zone before deploying workloads?"

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### **Your Defense**

What is a Landing Zone?

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What capabilities does it provide?

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How does it reduce future risk?

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### **Architecture Principle**

Strong foundations reduce future complexity.

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## **DEFENSE ROUND 5**

# **MULTI-ACCOUNT STRATEGY**

### **Executive Question**

"Why not keep everything inside one account?"

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### **Defend Multi-Account Architecture**

Security Benefits

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Cost Benefits

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Governance Benefits

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Compliance Benefits

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## Executive Summary

Why is account isolation considered an AWS best practice?

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## ◆ DEFENSE ROUND 6

### THE BLAST RADIUS CHALLENGE

#### Scenario

A developer accidentally deletes production resources.

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#### Architecture A

Single Account Model

Impact:

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#### Architecture B

Multi-Account Model

Impact:

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#### Defense Question

Which architecture better limits business risk?

Why?

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### **Architecture Principle**

Great architects design for failure before failure occurs.

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## **DEFENSE ROUND 7**

### **COST GOVERNANCE**

#### **Executive Question**

"How do we maintain financial control across multiple accounts?"

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#### **Your Defense**

Which AWS Organizations feature helps solve this challenge?

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How does it improve visibility?

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How does it improve governance?

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#### **Financial Architecture Principle**

Visibility creates accountability.

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## ◆ DEFENSE ROUND 8

# SECURITY GOVERNANCE

### Scenario

Developers request permission to:

- ✓ Disable CloudTrail
  - ✓ Disable GuardDuty
  - ✓ Remove Security Hub
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### Executive Question

Should these permissions be approved?

☐ YES

☐ NO

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### Defend Your Position

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### Governance Recommendation

Which SCP controls would you implement?

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## ◆ DEFENSE ROUND 9

# ENTERPRISE ACCOUNT MODEL

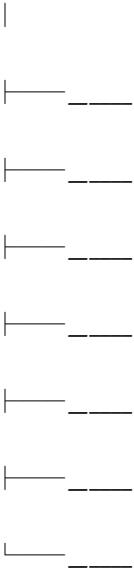
### Executive Board Request

Design the AWS account structure you would recommend.

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### Your Architecture

AWS Organization



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### Defense

Explain why each account exists.

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## ◆ DEFENSE ROUND 10

### EXECUTIVE ARCHITECT REVIEW

You are presenting to:

◆ Chief Executive Officer

◆ Chief Technology Officer

◆ Chief Information Security Officer

◆ Chief Financial Officer

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#### Final Executive Questions

1. Why is governance essential in AWS?

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1. Why should AWS Organizations be implemented?

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1. Why are SCPs critical?

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1. Why should workloads be separated?

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1. Why is a Landing Zone valuable?

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# ARCHITECT DECISION MATRIX

Rate your proposed architecture.

| Category               | Score (1-5) |
|------------------------|-------------|
| Security               | _____       |
| Governance             | _____       |
| Compliance             | _____       |
| Cost Management        | _____       |
| Scalability            | _____       |
| Resilience             | _____       |
| Operational Excellence | _____       |

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## Total Score

\_\_\_ / 35

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## Architecture Rating

30-35

✓ Enterprise Architecture Ready

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24-29

✓ Strong Architecture

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18-23

✓ Moderate Maturity

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Below 18

✓ Architecture Requires Improvement

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## ◆ EXECUTIVE APPROVAL RECOMMENDATION

As the Lead Cloud Governance Architect:

Would you approve this architecture for enterprise deployment?

☐ YES

☐ NO

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### Executive Summary

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## ◆ THINK LIKE A CLOUD GOVERNANCE ARCHITECT

Before approving any cloud environment ask:

- ◆ Is governance centralized?
- ◆ Are security controls protected?
- ◆ Is blast radius minimized?
- ◆ Is compliance enforceable?
- ◆ Can the environment scale safely?
- ◆ Are financial controls visible?
- ◆ Is operational ownership clear?

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## ◆ FINAL TAKEAWAY

Technology alone does not create enterprise readiness.

Governance transforms cloud infrastructure into a secure, scalable, and sustainable operating model.

The role of a Cloud Governance Architect is not merely to deploy resources.

It is to ensure that every resource exists within a framework of accountability, control, and trust.

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## **MZ UCA ARCHITECT PRINCIPLE**

**"Anyone can deploy cloud resources.**

**Architects design systems.**

**Governance Architects design control."**

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## **Zulu Mthulisi**

### **Cloud Governance Architect**

**MZ UCA Cloud Academy**

**Building Architects. Establishing Governance. Creating Leaders.**